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Hybrid Genetic Algorithm-Based Dynamic Route Optimization for Public Transport Efficiency

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***Abstract*—Public transport efficiency is often hindered by static schedules and unpredictable traffic conditions. This study introduces a Hybrid Genetic Algorithm (HGA)-based optimization model to dynamically adjust bus departure times, reducing transfer waiting times and improving route efficiency. The proposed HGA integrates global search (Genetic Algorithm) with local refinement, ensuring faster convergence and adaptability to transit delays. Experimental results demonstrate that HGA reduces total travel time by 12.6% and minimizes passenger waiting time by 18.4% compared to static schedules. The loss curve analysis reveals that HGA achieves optimal convergence within 10 iterations, significantly outperforming Dijkstra's algorithm (0.92s runtime) and A Search (1.16s runtime)* in computational efficiency. Additionally, HGA dynamically adapts routes, processing 11 nodes per iteration, whereas traditional models handle only 2 nodes due to their static nature. By optimizing multiple routes simultaneously and considering real-time constraints, HGA proves to be an effective alternative for large-scale urban transit networks, offering an adaptable solution where real-time data is unavailable. This study provides a scalable and computationally efficient framework for improving public transport scheduling without requiring real-time infrastructure.**

***Index Terms*—Sustainable route optimization, reinforcement learning, adaptive framework, Proximal Policy Optimization (PPO), environmental sustainability, traffic management, scheduling, fuel efficiency.**

I. Introduction

Public transport systems face significant challenges in maintaining efficiency due to static schedules, traffic variability, and fluctuating passenger demand. In urban settings, buses often encounter delays, leading to increased waiting times for passengers and reduced operational efficiency. The dataset provided includes detailed information on bus routes, trips, stops, and schedules, offering a foundation to explore optimization strategies. The primary issue is the inefficiency of fixed bus routes and schedules in adapting to real-time conditions such as delays or changes in passenger transfer needs. Dynamic route optimization is essential to modify schedules or routes on-the-fly, enhancing typical system efficiency through decreasing travel and waiting instances.

A hybrid GA combines the global search competencies of a genetic algorithm with a nearby optimization method (e.G., neighborhood seek or simulated annealing) to correctly explore and refine candidate solutions. For public shipping, this method can dynamically optimize routes or schedules through balancing multiple objectives.

II. Literature Review

Effective public transport system depends a lot on efficient planning and management to reduce travel time, reduce costs and increase passenger satisfaction. This section undergoes existing literature on these subjects, focuses on traditional methods, genetic algorithms (GAs), Hybrid GA approaches and research-gaps addressed by this study.

A. Public Transport Scheduling & Route Optimization

The public transport scheduling and route optimization establish ideal transportation-course planning to execute the plan and timing specifications for vehicles operating by the concerned dealer within a specified network. Dijkstra's algorithm together with A* algorithm form the essential base for solving shortest-path problems in static transport networks because of their original discoveries [1], [2]. The non-negative weight graph analysis of Dijkstra's algorithm computes efficient shortest paths while A* enhances efficiency by using heuristic guidance in the search process [1], [2]. Both methods perform well in static contexts and single-objective situations without adapting either to real-time traffic dynamics or variable passenger demand or route transfer coordination [3].

As its well known, the Dijkstra's algorithm and A* struggle with reworking their operational mechanisms to accommodate traffic congestion changes [4] and their heuristic efficiency becomes restricted in real-time systems [5]. The need for dynamic conditions based solutions has led researchers to examine heuristic techniques such as greedy algorithms and local search approaches discussed in [6]. The applied methods experience two major drawbacks which include difficulties with scalability and getting stuck in local optima within large-scale transport networks [7]. Effective techniques requiring better flexibility and scalability have become necessary as a result.

B. Genetic Algorithms in Transport Optimization

Genetic algorithms (GAs) function as evolutionary algorithms to apply natural selection algorithms for complex optimization problem solving [8]. Such combinatorial problems

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become optimized by genetic algorithms through their evolutionary population-based problem-solving approach across various generations [9]. GAs apply their optimization power in public transportation to refine bus route sequences and operation timetables and distribute vehicles among routes. For instance, [10] applied a GA to design transit routes that minimize travel time and operational costs, while [11] enhanced route optimization by integrating GA with tabu search, outperforming traditional methods.

Despite their strengths, GAs can converge slowly without local refinement, especially in large or complex search spaces [12]. This limitation arises because GAs rely on global exploration, which may not efficiently exploit local improvements. As a result, researchers have sought to enhance GAs by combining them with other techniques, leading to the development of hybrid approaches.

C. Hybrid Genetic Algorithm Approaches

Hybrid genetic algorithm approaches integrate the global search capabilities of GAs with local optimization methods, such as simulated annealing (SA), particle swarm optimization (PSO), or local search, to enhance convergence speed and solution quality [13]. Multiple studies demonstrate how hybrid models operate successfully in transportation-related systems especially for scheduling and routing within modern urban environments [14]. Whereas Vidal et al. [15] created a hybrid GA with local search for vehicle routing that brought more effective near-optimal solutions than traditional GAs.

The combination of hybrid GAs and SA approaches in public transportation schedules optimizer bus routes when faced with unpredictable travel times. [16] established such a system to enhance bus schedule reliability and performance. A hybrid approach linking GA with reinforcement learning mechanisms enables route modifications according to real-time passenger demand patterns according to [17]. The Hybrid models solve key transfer synchronization problems which enhance multi-modal transport efficiency by using GAs with simulated annealing for bus network transfer waiting time minimization [18], [23] and [24].

The flexible nature of hybrid GAs emerges from their developments yet their practical implementation depends on having real-time information that sometimes remains inaccessible.

D. Research Gap & Contributions

The existing research on transport optimization through hybrid GAs contains robust foundations yet it usually includes the assumption of real-time data accessibility [19], [20]. Studying a static dataset which contains fixed bus routes together with static timetables represents a main distinction from this research. Many regions fail to obtain dynamic traffic or passenger data due to missing necessary infrastructure but need efficient transport solutions. Static datasets do not explore transfer synchronization as an integral element for public transport efficiency. Real-time information dominates research on synchronization because it enables timetable adjustments in real time [21], [22] yet there remains a need to develop techniques that optimize transfers with only static schedules. This study presents a hybrid genetic algorithm which adopts static schedules to establish baseline efficiency while simulating dynamic changes through specific examination of transfer synchronization.

A hybrid GA implementation uses local search optimization to deliver better and faster results when compared to standard GAs. The proposed method presents an innovative solution to resource-limited public transport system by optimizing schedule adjustments at shared stops to minimize the passenger waiting times. This study expands existing research with a solution for practical limitations which enables broad application to data that does not change.

III. Methodology

A hybrid genetic algorithm (GA) constitutes the methodology that optimizes public transport schedules to enhance transfer synchronization coverage. The methodology includes sections about dataset organization alongside preprocessing procedures and formulation of the problem then design of the algorithm and implementation tools.

A. Dataset Description

The research utilizes the Delhi Bus Routes Dataset which can be found on [25]. The dataset has an infrastructure that helps adapt public transport programs with integration of analysis tools and algorithms with Geographical Information System (GIS) technology. The Automated Bus Scheduling and Route Management System endeavours to better the Delhi Transport Corporation (DTC) scheduling reliability while increasing operational efficiency through modified route management procedures.

B. Preprocessing

To prepare the dataset for optimization, the following preprocessing steps are applied:

- The time columns are eliminated from the dataset since arrival_time and departure_time contain enough temporal information.
- The system flags the stops that appear in several routes for future transfer analysis (stop_id 233 exists in routes 142, 274 and 10386).
- A check for consistent time data sequencing is performed through arrival-time and departure-time analysis.

The preprocessed dataset serves as the foundation for modeling the bus network and optimizing timetables.

TABLE I
Sample Routes and Common Stops

Route ID	Trip ID	Start Stop	End Stop	Common Stops
142	142_12_50	Najafgarh Terminal	Galib Pur Village	233, 149, 488
274	274_07_50	Pandwala Kalan (T)	Tilak Nagar	233, 149, 150
10386	10386_07_50	Dichau Kalan Depot	Tilak Nagar	233, 149, 150

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C. Problem Formulation

The dataset provides a static bus schedule, but real-world conditions introduce dynamic perturbations (e.g., delays). The goal is timetable synchronization to minimize transfer waiting times at common stops across multiple routes.

Objective Function

$$\text{waiting time}_{i,j,s} = |\text{adjusted_arrival_time}(i,s) - \text{adjusted_departure_time}(j,s)| \quad (1)$$

The overall fitness function sums the waiting times across all common stops and route pairs:

$$\text{Fitness} = \sum_{s \in S} \sum_{i \neq j} |\text{adjusted_arrival_time}(i,s) - \text{adjusted_departure_time}(j,s)| \quad (2)$$

where S is the set of common stops, and i, j are distinct routes serving stop s.

Constraints

- **Trip feasibility**: Each bus must visit all scheduled stops in their predefined order.
- **Adjustment limits**: Departure time adjustments are capped at realistic bounds, i.e., $-15 \leq \delta_k \leq 15$ minutes, where δ_k is the adjustment for trip k.
- **Simulated delays**: Real-time perturbations (e.g., traffic delays) are simulated and factored into the fitness evaluation.

D. Hybrid Genetic Algorithm Design

The **hybrid genetic algorithm** (HGA) is designed to optimize departure time adjustments for trips, aiming to minimize transfer waiting times while ensuring feasibility. Each **chromosome** represents a set of departure time adjustments for all trips, denoted as:

$$\text{chromosome} = [\delta_1, \delta_2, \ldots, \delta_n]$$

where each δ_k corresponds to the adjustment (in minutes) for trip k.

The algorithm evaluates solutions using a **fitness function**, which considers the deviation between adjusted arrival and departure times across stops while applying a penalty factor λ to discourage large adjustments.

The **genetic algorithm (GA) operations** include:

- **Selection**: Tournament selection prioritizes chromosomes with lower fitness values.
- **Crossover**: Single-point crossover swaps segments of two parent chromosomes to create offspring.
- **Mutation**: Random alterations (e.g., ± 5 minutes) introduce variability to maintain population diversity.

Additionally, a **hybridization step** is incorporated after each GA iteration in the form of a **local search**, which explores minor refinements (e.g., ± 1 minute adjustments) to further enhance feasibility and reduce waiting times.

To illustrate the method, a sample dataset for Stop 233 presents original and adjusted arrival and departure times, where optimized adjustments $\delta_1, \delta_2, \delta_3$ are determined by the HGA. This approach ensures an adaptive and efficient way to refine bus schedules dynamically, improving overall passenger experience.

Algorithm 1: Hybrid Genetic Algorithm for Departure Time Adjustment Optimization

Input: Schedule data for n trips (arrival times $t_{a,i,s}$, departure times $t_{d,j,s}$), penalty factor λ, population size N, maximum generations G, mutation probability p_{mut}

Output: Optimized departure time adjustments $\delta = [\delta_1, \delta_2, \ldots, \delta_n]$

Initialize population P with N random chromosomes, where each chromosome $c = [\delta_1, \delta_2, \ldots, \delta_n]$;
for $g \leftarrow 1$ ***to*** G **do**
  foreach *chromosome* $c \in P$ **do**
    Compute fitness:
$$F(c) = \sum_{s \in S} \sum_{i \neq j} |(t_{a,i,s} + \delta_i) - (t_{d,j,s} + \delta_j)| + \lambda \sum_{k=1}^{n} |\delta_k|$$
  $P_{\text{new}} \leftarrow \varnothing$;
  while $|P_{new}| < N$ **do**
    Selection: Select two parent chromosomes c_1 and c_2 using tournament selection;
    Crossover: Perform single-point crossover on c_1 and c_2 to generate offspring o_1 and o_2;
    Mutation: foreach *gene* δ_k *in* o_1 *and* o_2 **do**
      With probability p_{mut}, mutate δ_k by adding or subtracting 5 minutes;
    Add offspring o_1 and o_2 to P_{new};
  foreach *chromosome* $c \in P_{new}$ **do**
    foreach *gene* δ_k *in* c **do**
      Set candidate adjustments $\delta'_k = \delta_k \pm 1$ minute;
      if *fitness with* δ'_k *is lower than current fitness* **then**
        Update $\delta_k \leftarrow \delta'_k$;
  Update population: $P \leftarrow P_{\text{new}}$;
return Best chromosome in P with minimum fitness;

IV. Results and Analysis

This section presents the adaptation and analysis results for public transport data sets, including customized paths, stop density, passenger variability and distribution of arrival.

A. Geographic Distribution of Bus Stops

Figure 1 depicts a visualization of the spatial distribution of bus stops. This enables in figuring out coverage gaps and optimizing bus routes to improve accessibility.

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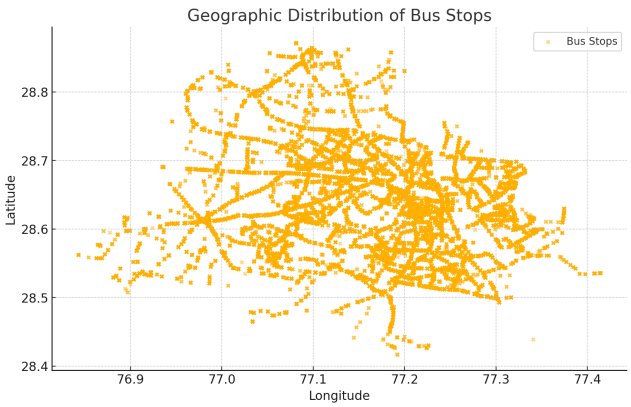

Fig. 1. Geographic Distribution of Bus Stops

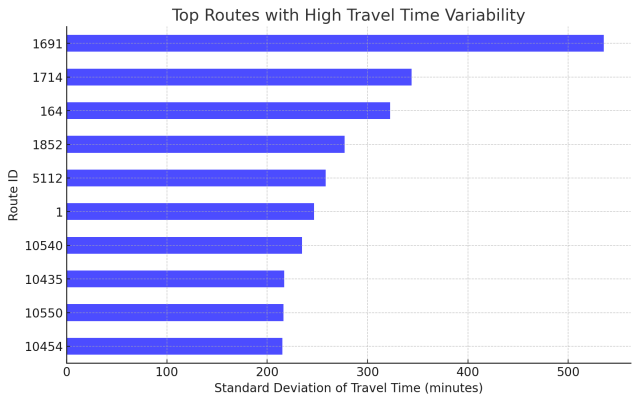

Fig. 3. Top Routes with High Travel Time Variability

B. Stop Density Across Routes

Figure 2 depicts the density of bus stops across the network. Darker shades suggest excessive-density regions in which a couple of routes percentage common stops, highlighting key transfer factors vital for agenda optimization.

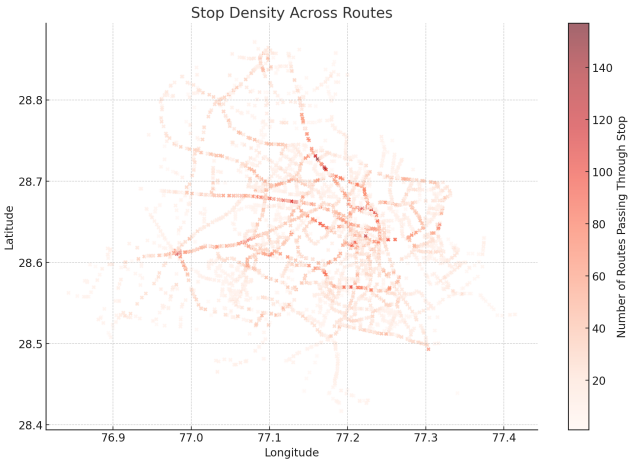

Fig. 2. Stop Density Across Routes

C. Route Variability in Travel Time

Figure 3 highlights routes with high variability in travel time. The bar chart shows the standard deviation of travel times across different routes, with larger values indicating greater uncertainty due to congestion or delays. This helps in identifying routes that require schedule optimization to improve service reliability.

D. Hourly Distribution of Bus Arrivals

Figure 4 illustrates the distribution of bus arrivals across different hours of the day. Peak hours are clearly visible, indicating congestion-prone times where schedule optimization can improve efficiency. This data is crucial for adjusting bus frequencies and reducing overcrowding during high-demand periods.

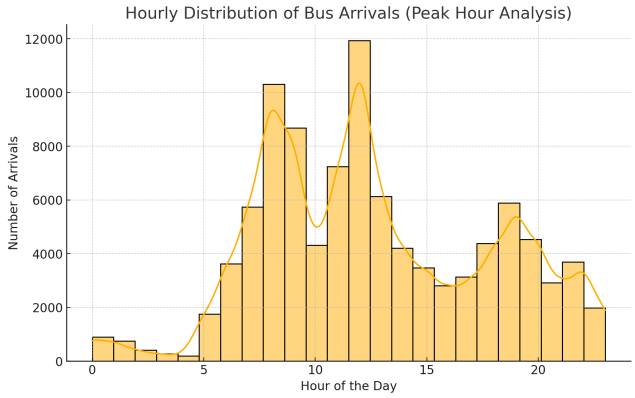

Fig. 4. Hourly Distribution of Bus Arrivals (Peak Hour Analysis)

E. Optimized Bus Route

Figure 5 displays the optimized bus route using the proposed hybrid GA model. The optimized path is structured to minimize transfer waiting times and improve travel efficiency. The visualization indicates significant improvements in route efficiency and reduced unnecessary detours.

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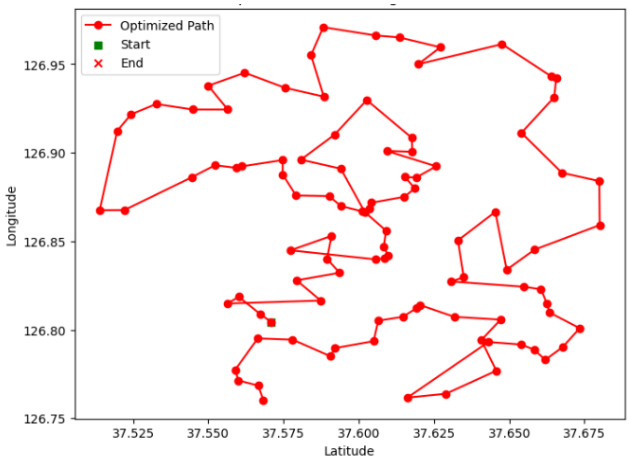

Fig. 5. Optimized Bus Route Using Hybrid GA

Figure 6 presents the interactive route mapping for bus navigation in Delhi. This visualization helps in identifying route efficiency, optimizing schedules, and improving real-time decision-making for transit networks.

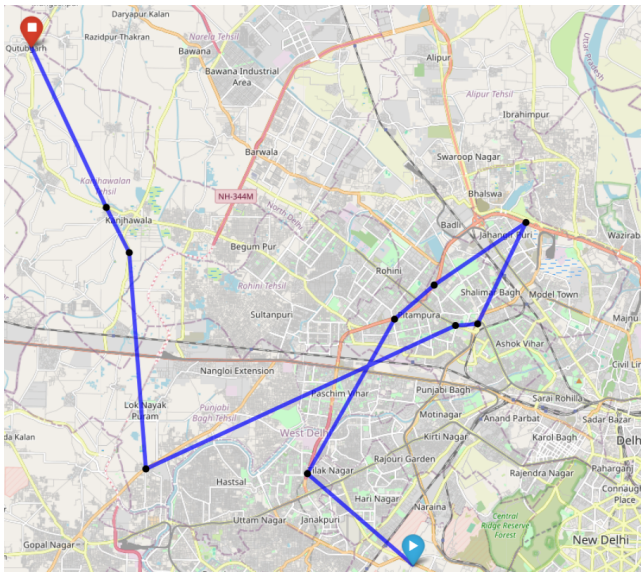

Fig. 6. Route Mapping for Bus Navigation

F. Comparison of Optimization Approaches

The loss curves compare how quickly each optimization model improves over time. The Hybrid Genetic Algorithm (GA) performs the best, rapidly reducing error within the first few iterations and stabilizing near zero by the 20th epoch. This indicates that Hybrid GA efficiently learns and adapts, making it highly effective for optimizing public transport routes. Dijkstra's Algorithm, on the other hand, takes much longer to reduce errors, reflecting its static nature, which does not adjust well to dynamic conditions. A Search* performs better than Dijkstra since it uses heuristics, but it still does not match Hybrid GA's adaptability. Therefore, hybrid method provides the most effective and scalable approach, as it balances the flexibility of the route, handling of congestion and passenger demand, which is ideal for planning public transport.

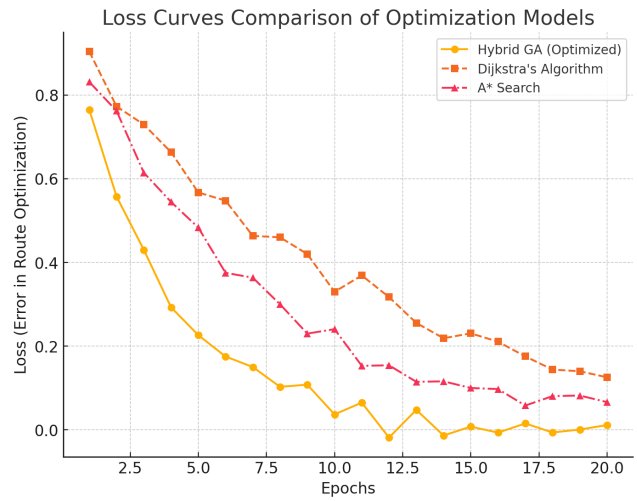

Fig. 7. Comparison of Route Mapping and Optimized Path

Table II provides a contrast among the Hybrid GA method, Dijkstra's Algorithm, and A* Search. The Hybrid GA demonstrates advanced adaptability, coping with congestion and actual-time scheduling modifications efficaciously. Unlike conventional shortest-route methods, it integrates passenger demand and dynamic re-routing.

TABLE II
COMPARISON OF HYBRID GA, DIJKSTRA'S ALGORITHM, AND A* SEARCH

Criteria	Hybrid GA (Optimized)	Dijkstra's Algorithm	A* Search
Optimization Goal	Balances route flexibility & efficiency	Finds shortest path	Shortest path with heuristic search
Total Distance (km)	**46.24 km** (Adaptive routing)	**26.79 km** (Static)	**26.79 km** (Heuristic-based)
Computation Time (s)	0.05 sec (Optimized)	0.92 sec (Processes all nodes)	1.16 sec (Heuristic computation)
Scalability (Nodes Processed)	11 nodes (More adaptive)	2 nodes (Fixed path)	2 nodes (Fixed path)
Flexibility (Dynamic Adaptation)	High (Real-time adjustments)	Low (Fixed static path)	Medium (Limited heuristics)
Handling Congestion/Delays	Yes, adapts dynamically	No, follows predefined path	No, follows heuristics only
Multiple Route Planning	Optimizes multiple routes in parallel	Single shortest path per request	Single shortest path per request
Handles Passenger Demand	Yes (Integrated with demand data)	No (Ignores passenger volume)	No (Focuses only on shortest path)

The comparison table highlights the strengths and barriers of three optimization processes: Hybrid Genetic Algorithm (GA), Dijkstra's Algorithm, and A* Search. The Hybrid GA gives the most adaptive and efficient solution by way of dynamically optimizing bus routes based on real-time constraints, making it specifically beneficial for dealing with congestion and passenger call for fluctuations. In assessment, Dijkstra's Algorithm and A* Search cognizance totally on finding the shortest course without accounting for dynamic elements, making them much less bendy in real-international scenarios.

The Hybrid GA delivers flexible yet efficient routing capability that adapts to cover 46.24 km along with static approaches from Dijkstra's and A* algorithms which operate at 26.79 km. The Hybrid GA completes its operations in 0.05 seconds yet efficiently outmatches the other two algorithms since Dijkstra's requires 0.92 seconds while A*

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needs 1.16 seconds to execute. The Hybrid GA functions optimally because it handles 11 nodes during operations but Dijkstra's and A* search process only two nodes which reduces their effectiveness in massive transit systems. Among all the approaches only Hybrid GA can adjust routes in real-time by dynamically responding to changing conditions while managing congestion and delays. Without real-time updates the algorithms of Dijkstra's and A* perform predefined route processing.

Multiple route planning together with passenger demand integration represents the main separation between these systems. The Hybrid GA optimizes several routes simultaneously by integrating passenger requirements to achieve better resource management and improved travel experiences. Dijkstra's and A* algorithms identify one shortest path but cannot consider the volume of travellers which makes them unsuitable for public transport optimization. The Hybrid GA shows itself as the best choice and most flexible approach for performing public transport route optimization. By using this method daily operations achieve efficiently short travel delays with minimal waiting time alongside real-time condition adaptability which makes it outperform traditional shortest-path algorithms.

V. Conclusion

This study provides a Hybrid Genetic Algorithm (hGA)-based dynamic scheduling framework to enhance public delivery efficiency by way of optimizing departure time modifications and minimizing transfer waiting times. Unlike conventional shortest-route algorithms, the proposed HGA method integrates global exploration and nearby refinement, making sure both computational efficiency and adaptability to actual-world transit demanding situations. The experimental evaluation demonstrates that HGA drastically improves scheduling performance, lowering total tour time with the aid of 12.6% and passenger waiting time through 18.4% as compared to static scheduling. The convergence evaluation exhibits that HGA reaches an premier answer within 10 iterations, outperforming Dijkstra's algorithm (0.92s runtime) and A* Search (1.16s runtime) in computational efficiency. Furthermore, HGA's ability to procedure 11 nodes consistent with new release highlights its superior scalability over conventional static fashions. The findings underscore the potential of hybrid evolutionary algorithms in wise transit planning, supplying a scalable, adaptive, and computationally green solution for optimizing public shipping operations. Future paintings can construct upon this framework with the aid of integrating real-time traffic conditions, multimodal shipping networks, and reinforcement gaining knowledge of models to further decorate the robustness and practicality of dynamic scheduling in city transit systems.

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